Bus Stop Check

Design Specification

Design and evaluation of a local app
for manual OSM bus-stop facility checks

Design Studio · WS 2026/27
Revised draft v5 · 2 November 2026

Project focus and design objective

Main focus

Design and evaluate a local app for manual checks of OSM bus-stop facilities. Assess workflow clarity and correct recording of uncertainty.

Design question

How should the interface guide users through facility checks and help them distinguish missing OSM records from unclear imagery?

Role of OSM data quality

Better data quality motivates the project. A measurable improvement to the OSM database is outside this project's evaluation scope.

Intended users and project scope

Intended users

OSM contributors reviewing passenger waiting areas.
Community comments received; task-review participation remains unconfirmed.

Four facilities

Shelter · Bench · Lighting · Stop sign

Working boundary

Single user, local computer, up to 50 OSM objects per list.
Human interpretation; no automatic OSM edits or repair decisions.

Community feedback and design decisions

Feedback theme	Design response
Source restrictions	Removed Google entries from new checks; use sources suitable for OSM.
Limited aerial detail	Retained Cannot judge; an unclear view does not establish facility absence.
Alternative imagery	Deferred basemap.at integration; street imagery remains the focus.

Existing base: OSM import, manual review, local saving and seven-column export.
Evidence: OSM Austria topic 149099, posts 2–7 (21–22 Oct 2026); three contributors.

Methods across four design stages

01 Analysis (Analyse)

Group received forum comments and tool documents by theme.
Link each design decision to its source, reason and follow-up.

02 Concept design (Konzeption)

User stories and desktop / optional mobile sketches.
Expose unclear steps before changing the software.

03 Development (Entwicklung)

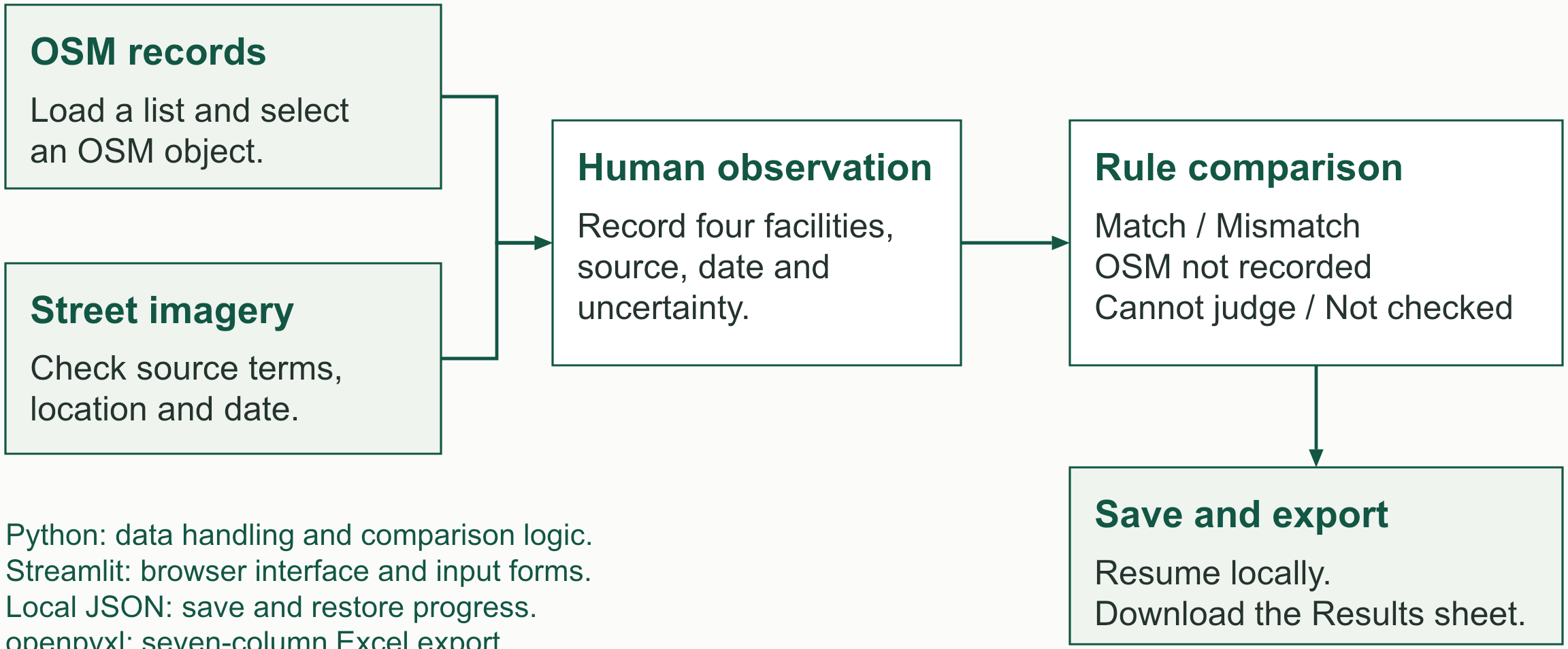
Refine source entries in the existing Python / Streamlit Demo.
Check local saving, restored provenance and seven-column export.

04 Evaluation (Evaluierung)

Planned task observation with think-aloud comments.
Find misunderstandings and errors to guide revisions.

Adapted from 2_Consulting.pdf, pp. 4 and 6. Methods selected for this project.

Review workflow and data flow



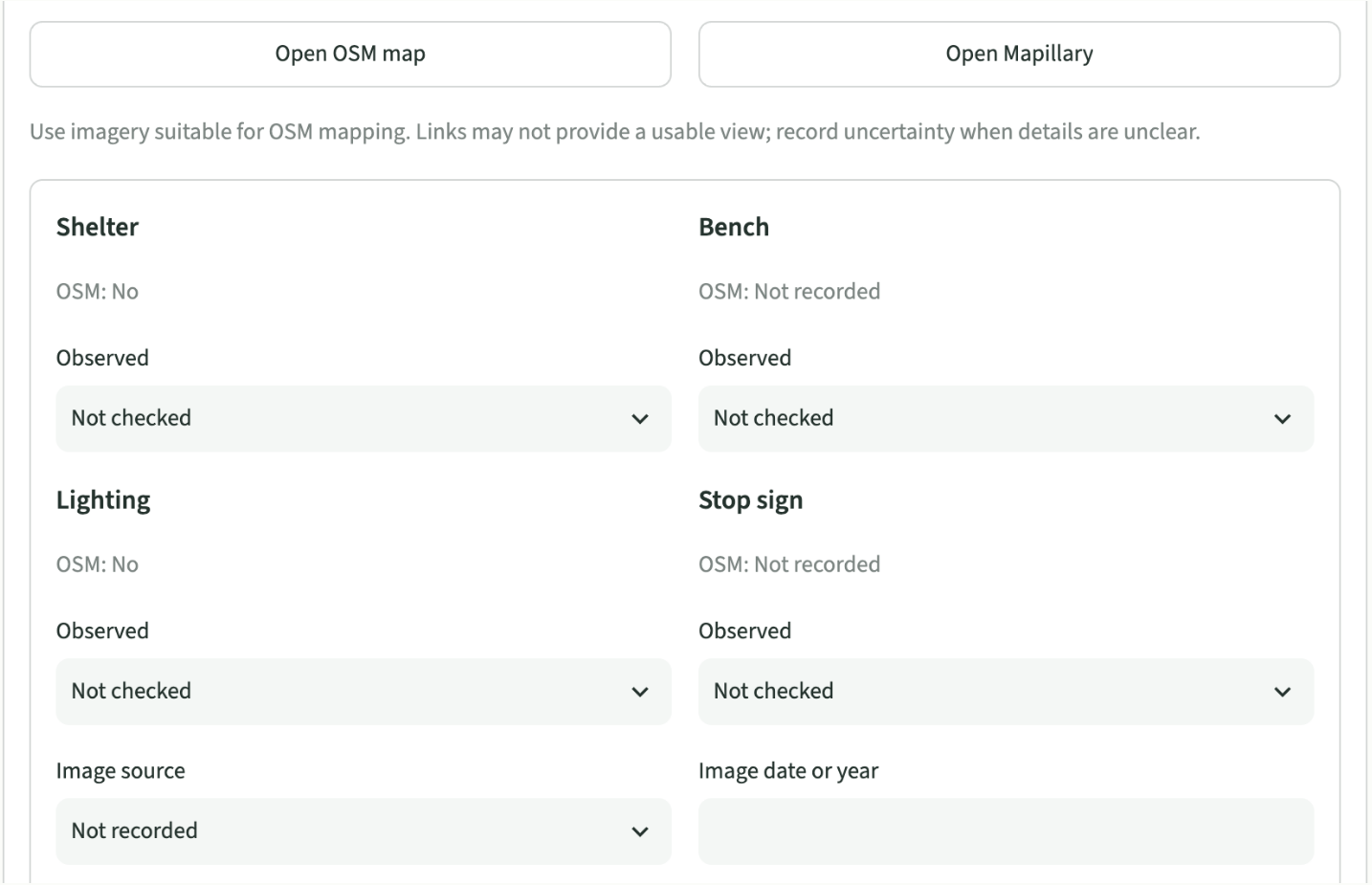
Python: data handling and comparison logic.
Streamlit: browser interface and input forms.
Local JSON: save and restore progress.
openpyxl: seven-column Excel export.

Observation interface after feedback

OSM record and human observation stay separate

Record uncertain or unavailable imagery without forcing yes/no.

Google entries removed from new checks; saved sources retained.



Comparison rules

OSM record	Human observation	Result
Yes	Yes	Match
Yes	No	Mismatch
Missing	Yes or No	OSM not recorded
Any value	Unclear / no imagery / outdated imagery	Cannot judge
Any value	Not checked	Not checked
Other than Yes/No	Yes or No	Cannot judge

Only explicit Yes/No values are compared. Uncertain observations take priority over missing tags.

Project output: one comparison sheet

One row = one facility for one OSM object

OSM object	Stop name	Facility	OSM record	Observed	Result	Notes
node/1005	Border Stop Example	Shelter	No	No	Match	Illustrative
node/1005	Border Stop Example	Bench	Not recorded	Yes	OSM not recorded	Illustrative
node/1005	Border Stop Example	Lighting	No	Yes	Mismatch	Illustrative
node/1005	Border Stop Example	Stop sign	Not recorded	No imagery	Cannot judge	Illustrative

Illustrative rows with hypothetical observations; these are not real verification results.

Additional outputs: local Demo, brief user instructions and documented evaluation.

Evaluation evidence and design changes

Community input received

Three contributors, six forum replies: source restrictions and imagery suitability.
Based on the description and screenshot; not an observed usability test.

Analyse and document decisions

Group comments by issue; record adoption or deferral with reasons.
Implemented: Google entry removal. Deferred: aerial imagery integration.

Planned user evaluation

Task: select a stop, record uncertainty, save, reopen and export.
Observe completion, errors, help needed and comments; refine and recheck.

Functional checks: 14 automated tests passed on 2 November 2026.
User-review participants and format remain subject to supervisor confirmation.